The *cis* and the *trans* faces of the organelle are entirely different, but interconnected. The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell. Materials to be packaged in the form of vesicles from the ER fuse with the *cis* face of the golgi apparatus and move towards the maturing face. This explains, why the golgi apparatus remains in close association with the endoplasmic reticulum. A number of proteins synthesized by ribosomes on the endoplasmic reticulum are modified in the cisternae of the golgi apparatus before they are released from its *trans* face. Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

Functions of Golgi apparatus

- Golgi apparatus helps in protein sorting from one compartment to another by the secretory pathway.
- Covalent modifications of proteins involving the addition of small sugar molecules occur in the ER and Golgi apparatus.

1.4.7.Ribosomes:

Ribosomes are the granular structures first observed under the electron microscope as dense particles by George Palade (1953). In the word ribosome, the pharse 'ribo' is derived from ribonucleic acid and 'somes' from the Greek word 'soma' which means 'body'. Ribosomes are tiny particles about 200 Å. They are composed of ribonucleic acid (RNA) and proteins. Ribosomes are not considered as organelles because of the lack of a membrane around them. However, when they are producing certain proteins they can become bound to the endoplasmic reticulum membrane. Free floating ribosomes are also present. Ribosomes are composed of both RNA and proteins. About 37 - 62% of ribosomes are made up of RNA and the rest is proteins. There are two types of ribosomes based on their sedimentation properties. Prokaryotes possess 70 S ribosomes and Eukaryotes possess 80 S ribosomes. The subunits of ribosomes are named owning to their sedimentation rate measured as special Svedberg Unit ('S'). The ribosomes share a core structure which is similar to all ribosomes despite differences in their size. The ribosomes are made up of two subunits - a small and a large subunit. The small subunit reads the mRNA while the large subunit joins the amino acids to form a chain of polypeptides.

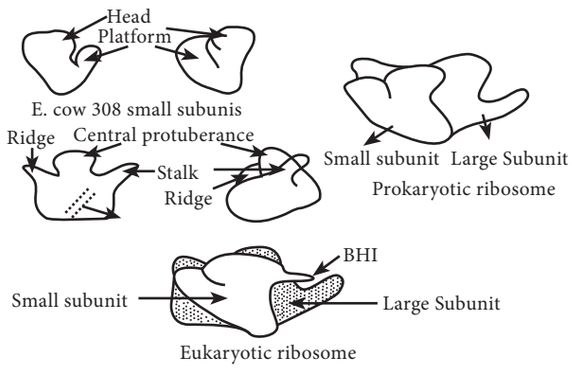

Figure 1.10 A schematic diagram of larger and smaller subunits of a ribosome

Functions of ribosomes:

- The bound and the free ribosomes are similar in structure and are involved in protein synthesis.

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